

Handwritten Digit Recognizer

A beginner-friendly AI/ML project in Python using Keras & TensorFlow

Overview

This project trains a neural network to recognize handwritten digits (0–9) using the classic MNIST dataset. It's a great first machine learning project because the dataset is built into Keras, the code is short, training takes just a couple of minutes on a laptop, and the results are easy to understand and satisfying to see in action.

Project Steps

- 1 Setup – install TensorFlow (includes Keras)
- 2 Load data – MNIST comes built into Keras, no downloading needed
- 3 Build a simple neural network – a few dense layers
- 4 Train – takes just a couple of minutes on a laptop
- 5 Test – check accuracy, try it on sample digits

Starter Code

Run this in a Jupyter notebook or plain .py file. You should hit around 97% test accuracy.

```
import tensorflow as tf
from tensorflow import keras

# Load the data (60,000 training images, 10,000 test images)
(x_train, y_train), (x_test, y_test) = keras.datasets.mnist.load_data()

# Normalize pixel values to 0-1 range
x_train = x_train / 255.0
x_test = x_test / 255.0

# Build a simple model
model = keras.Sequential([
    keras.layers.Flatten(input_shape=(28, 28)), # 28x28 image -> 784 numbers
    keras.layers.Dense(128, activation='relu'), # hidden layer
    keras.layers.Dense(10, activation='softmax') # output: 10 digits (0-9)
])

model.compile(optimizer='adam',
              loss='sparse_categorical_crossentropy',
              metrics=['accuracy'])

# Train
model.fit(x_train, y_train, epochs=5)

# Evaluate
test_loss, test_acc = model.evaluate(x_test, y_test)
print(f"Test accuracy: {test_acc:.2%}")
```

My Results

Achieved 97.7% Accuracy

Next Steps

- Test the model on your own handwriting (draw a digit, save as a 28x28 grayscale image, and feed it in)
- Try adding more layers or neurons to see how accuracy changes
- Visualize misclassified digits to understand where the model struggles
- Swap the dense layers for a convolutional neural network (CNN) for higher accuracy